

Pricing illiquidity in Strategic Asset Allocation: a calibration framework for asset allocation practitioners

Draft

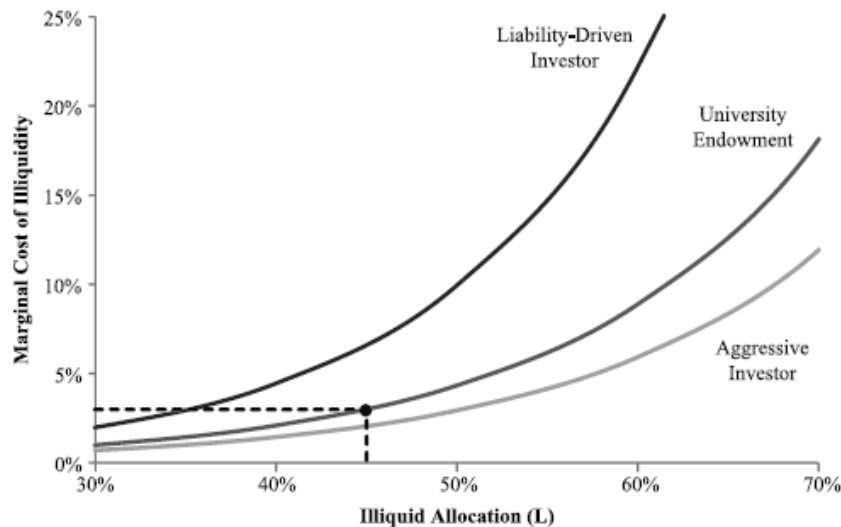
Agenda

1. Overview
2. Asset Classes & Illiquidity Levels
3. The Marginal Penalty Function
4. Portfolio Optimization with Illiquidity
5. Efficient Frontiers & Portfolio Composition
6. Simulation, Aggregation & Fitting
7. Back to Portfolio Optimization
8. A different Institutional Profile
9. What's next?

Overview

Bridging the Gap Between Theory and Calibration

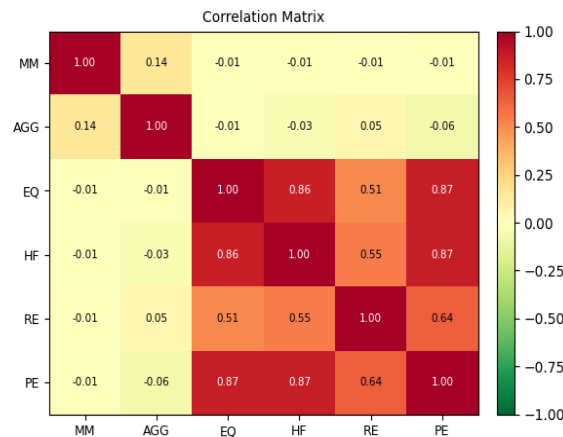
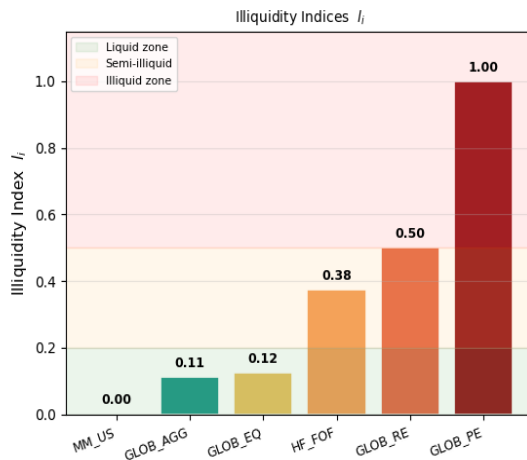
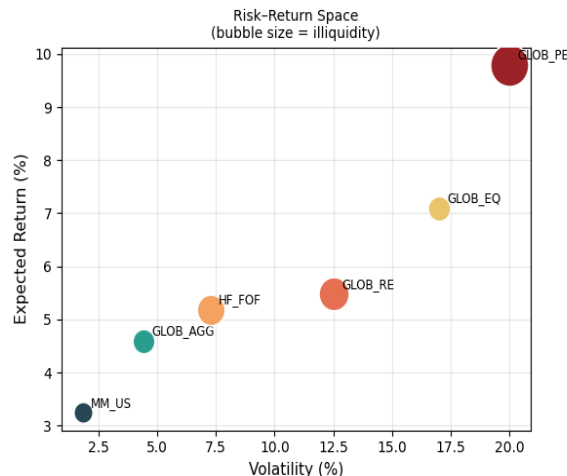
- Hayes, Primbs & Chiquoine (2015) extend the Markowitz framework by incorporating a marginal illiquidity penalty $c(L)$ into the MVO objective - but leave its identification to **subjective judgement**
- The **open question**: How do we estimate $c(L)$ rigorously from portfolio fundamentals?
- Our **approach**: simulate realistic portfolio dynamics under lock-ups, payouts and rebalancing constraints - then fit $c(L) = \alpha L^\beta$ directly from the simulated costs



Asset Classes & Illiquidity Levels

Defining the Investment Universe

Asset Class Overview



- Each asset class i is assigned an **illiquidity level** $l_i \in [0, 1]$, where $l_i = 0$ denotes a perfectly liquid asset (e.g., Fixed Income) and $l_i = 1$ denotes a fully illiquid asset (e.g., Private Equity)
- The portfolio illiquidity level is defined as: $L = \sum_{i=1}^n w_i \cdot l_i$, where w_i are portfolio weights and $\sum_{i=1}^n w_i = 1$

The Marginal Penalty Function

How different marginal curves behave with identical parameters ($\alpha = 0.05$, $\beta = 1.6$)

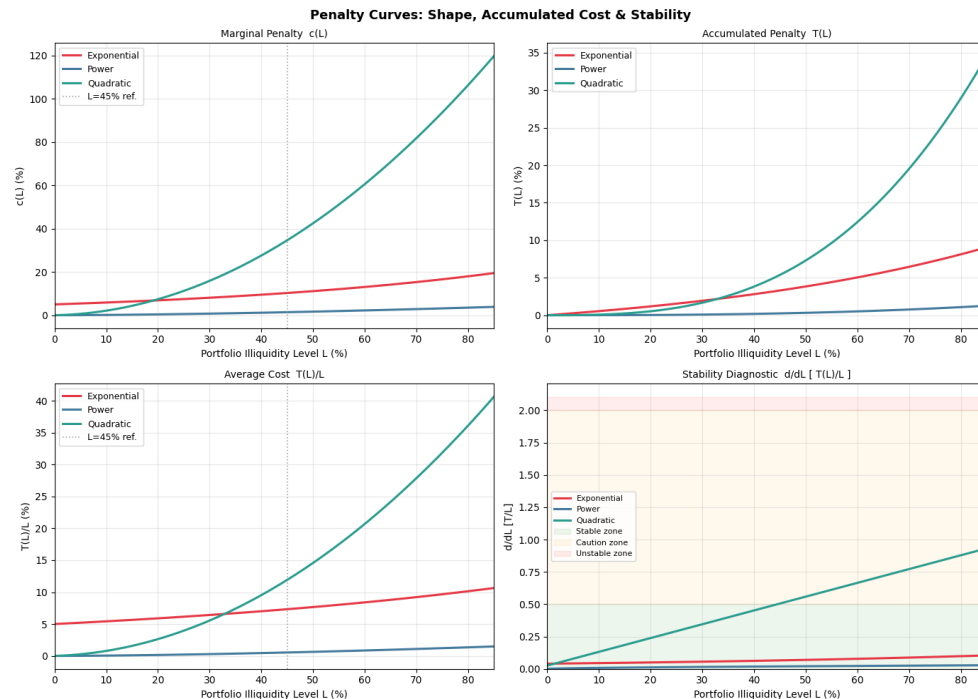
- Functional forms:**

1. Power: $c(L) = \alpha L^\beta$
2. Quadratic: $c(L) = \alpha L + \beta L^2$
3. Exponential: $c(L) = \alpha e^{\beta L}$

- Stability Diagnostic:** large values imply that small changes in portfolio weights - which shift L - cause large swings in illiquidity-adjusted returns

$$\hat{\mu} = \mu - l_i \frac{T(L)}{L}, \text{ potentially destabilising the optimizer}$$

- Economic interpretation (Power):** alpha determines the overall severity of the illiquidity penalty - how costly illiquidity is for that specific investor - while **beta** controls the shape of the penalty curve: a low beta reflects an investor who incurs costs uniformly across all illiquidity levels, whereas a high beta reflects an investor who tolerates illiquidity up to a threshold but faces sharply escalating costs beyond



Portfolio Optimization with Illiquidity

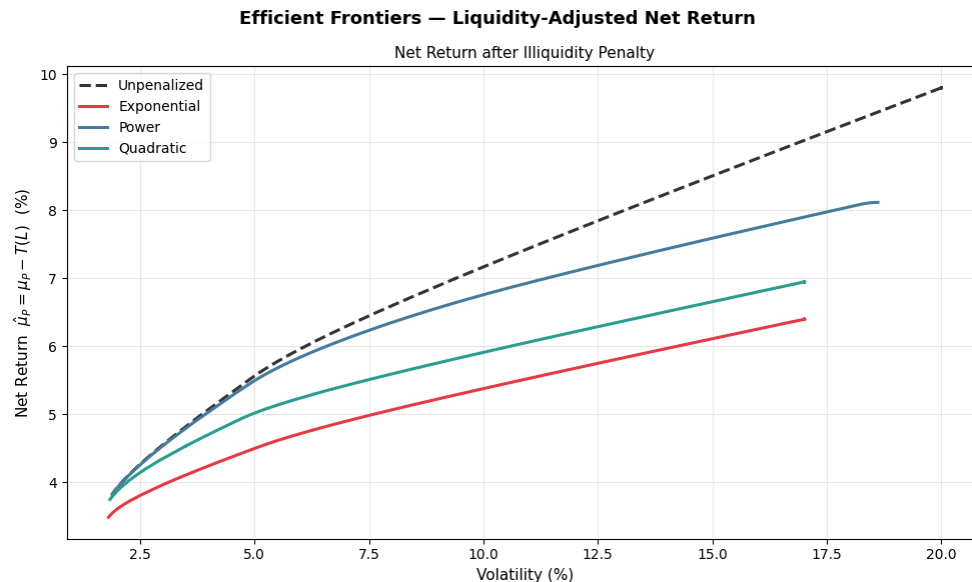
Net Efficient Frontiers ($\alpha = 0.05$, $\beta = 1.6$)

- Amend the objective by subtracting the accumulated illiquidity penalty:

$$\max_w w^T \mu - \frac{\gamma}{2} w^T \Sigma w - T(L(w)),$$

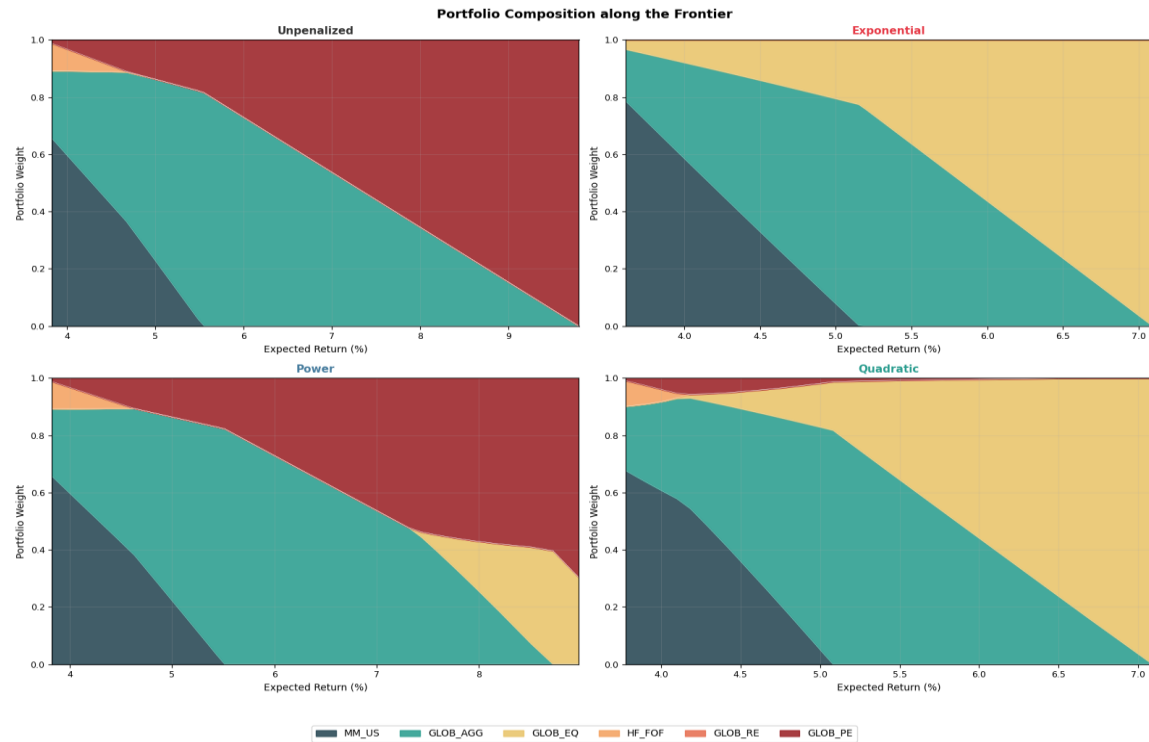
where $L = \sum_{i=1}^n w_i \cdot l_i$ and $T(L) = \int_0^L c(x) dx$

- As gamma approaches zero, the objective becomes $w^T \mu - T(L(w))$. The optimizer finds the mix that maximizes return minus the penalty and “stops” there. Lowering gamma further does not change the solution. The point where it stops depends directly on how aggressive the penalty is.



Efficient Frontiers & Portfolio Composition

Portfolio Composition by L ($\alpha = 0.05$, $\beta = 1.6$)



- The penalty reduces PE allocation and compresses the achievable return range, with GLOB_EQ emerging as the preferred liquid substitute given its high correlation with PE ($\rho = 0.87$) at a fraction of the illiquidity cost.
- Only the Power penalty delivers stable, economically consistent allocations - Exponential collapses the frontier to an implausibly narrow return range, while Quadratic shows erratic composition spikes - confirming Power as the appropriate functional form.

Simulation, Aggregation & Fitting

Simulation Design (1/2)

- We estimate $c(L)$ **empirically** by simulating realistic portfolio dynamics. For each target illiquidity level L , we run $N = 1000$ Monte Carlo scenarios over a 20-year horizon and measure the average annual **return foregone** by operating under lock-up constraints relative to an **unconstrained benchmark**.
- The framework is parameterised by an **annual payout ratio** δ representing the fraction of AUM distributed each year, and by a **minimum liquidity buffer** w_{min}^{liq} , always ensuring a floor on freely tradable assets. Both parameters are **adaptable to different institutional profiles** (e.g., pension fund benefit payments, endowment spending rules, or family office withdrawals).
- **Tranche system**: every asset purchase is recorded as a tranche with attributes asset, weight and bought_at. A tranche is locked if age = t - bought_at < lockup_years. This allows precise tracking of which positions are freely tradable and which are not, even when the same asset was purchased in different years.

Symbol	Value	Description
T	20 yr	Simulation horizon
N	1000	Monte Carlo scenarios
δ	10%	Annual payout ratio
w_{min}^{liq}	40%	Minimum liquid buffer
λ_{max}	8	Maximum lock-up period
L	[10%, 70%]	Illiquidity levels tested

Simulation, Aggregation & Fitting

Simulation Design (2/2)

- Each illiquid asset class is subject to a **lock-up period** during which positions cannot be freely liquidated: HF_FOF carries a 2-year lock-up, GLOB_RE 5 years, and GLOB_PE 8 years. Within the simulation, these constraints are enforced through a tranche-tracking system that records the purchase date of each position. Any attempt to reduce a locked position triggers a secondary market sale with the corresponding haircut cost.
- When a locked position must be liquidated before expiry — either to meet the annual payout or to restore the minimum liquidity buffer — the fund incurs an **asset-specific haircut** reflecting secondary market transaction costs. Haircuts are held constant throughout the simulation.
- The cost captures **two sources of friction: rebalancing gap** from lock-up constraints, and the **haircuts** on locked assets sold to restore the liquidity buffer and on the ones in the proportional payout. Simulated costs are used to fit $T(L) = \frac{\alpha}{\beta+1} L^{\beta+1}$ directly.

Asset	Haircut h_i
Fixed Income	0.1%
Public Equity	0.5%
Hedge Funds	8.0%
Real Estate	12.0%
Private equity	20.0%

Simulation, Aggregation & Fitting

The Annual Simulation Cycle

Step A – Market Returns

$$r_t \sim N(\mu, \Sigma)$$

Same realization shared by both **benchmark and actual** portfolio

Ensures: any **return difference captures** only the effect of **illiquidity constraints**, not different market outcomes

Step B – Benchmark

Portfolio with **no illiquidity constraints, resets to w_{target}** every year

Weights drift with the market:

$$w_{bd}^i = \frac{w_i^{target} \cdot (1 + r_i)}{\sum_j w_j^{target} \cdot (1 + r_j)}$$

Return computed on post-drift weights:

$$r_{bench} = w_{bd} \cdot r_t$$

Step C – Actual

$$\text{Drift: } w_i^{new} = \frac{w_i^{target} \cdot (1 + r_i)}{\sum_j w_j^{target} \cdot (1 + r_j)}$$

Proportional payout ($\delta = 5\%$): free assets → direct sale; locked assets → secondary market haircut

Liquidity buffer restore: first unlocked illiquid assets (no haircut), then locked (with haircut)

Constrained rebalancing:

$$\begin{aligned} \min_w \quad & \sum_i (w_i - w_i^{target})^2 \\ \text{s. t. } \quad & w_i \geq w_i^{locked} \end{aligned}$$

Step D – Annual Cost

Cost is measured on weights after drift, payout and restore - before year-end rebalancing:

$$\begin{aligned} \text{cost}_t = \\ (r_{bench} - r_{actual}) \\ + \text{cost}_h + \text{cost}_{fp} \end{aligned}$$

Step E – Fitting

Direct **estimate of the accumulated penalty** (non-linear OLS):

$$\hat{T} = A \cdot L^B$$

Parameters recovered analytically:

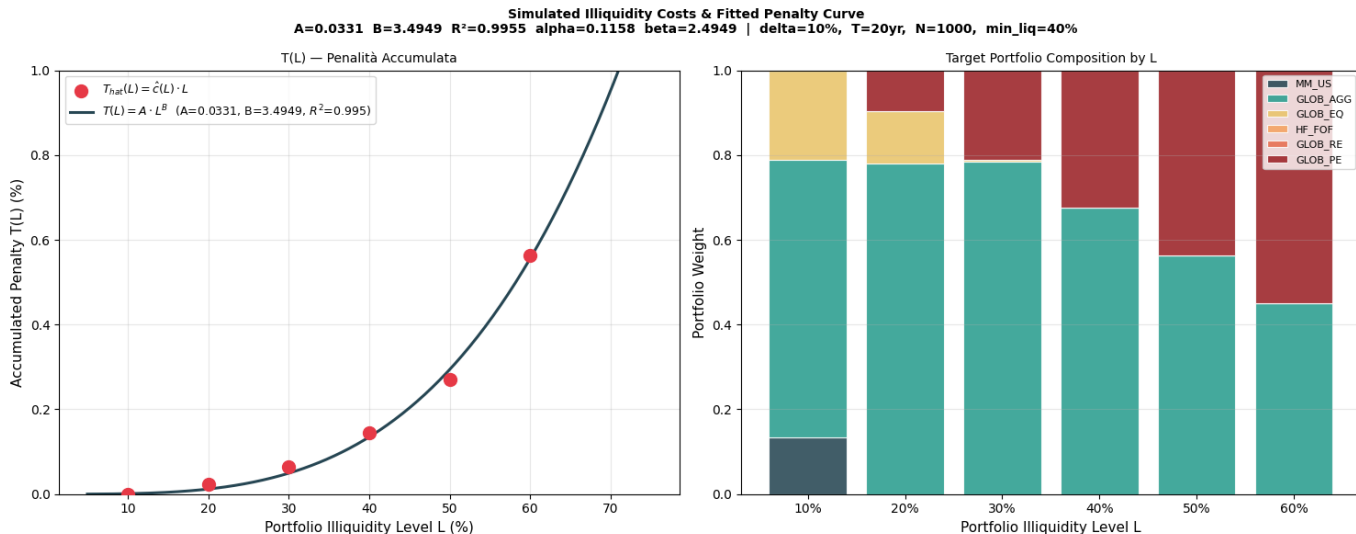
$$\beta = B - 1, \quad \alpha = A \cdot B$$

- Steps A through D are repeated for each of the T years, across all N scenarios

Simulation, Aggregation & Fitting

Fitting the parameters: pension fund ($\alpha = 0.12$, $\beta = 2.49$)

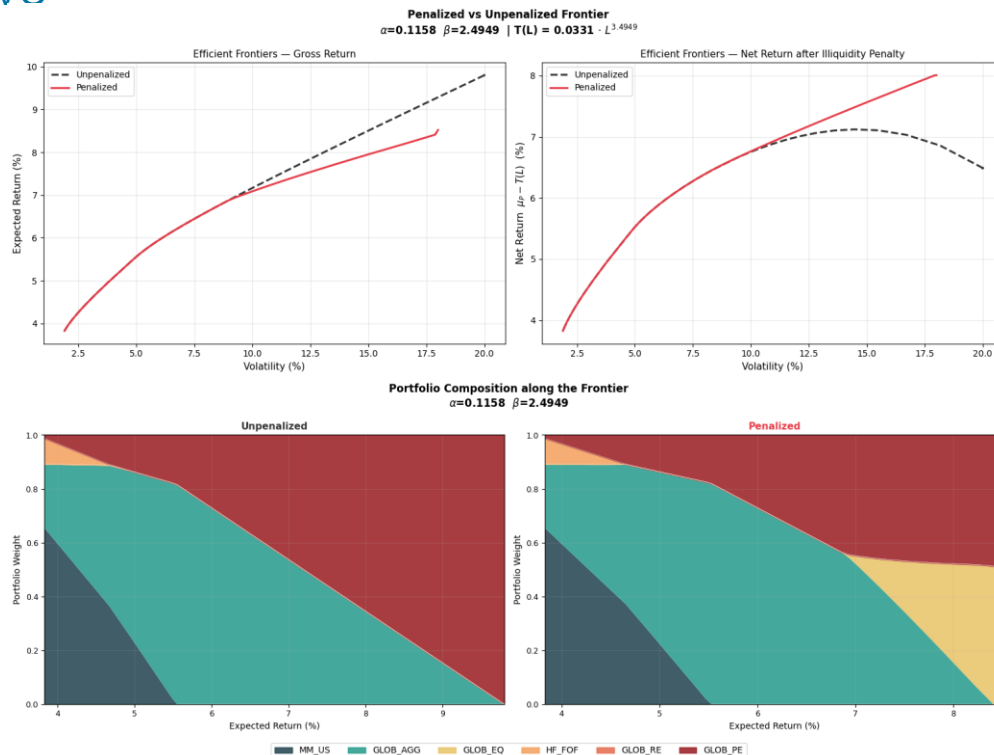
- The fitted parameters α and β provide a structured, simulation-based foundation for the accumulated penalty entering the penalized MVO: $T(L) = \frac{\hat{\alpha}}{\hat{\beta}+1} L^{\hat{\beta}+1}$



Back to Portfolio Optimization

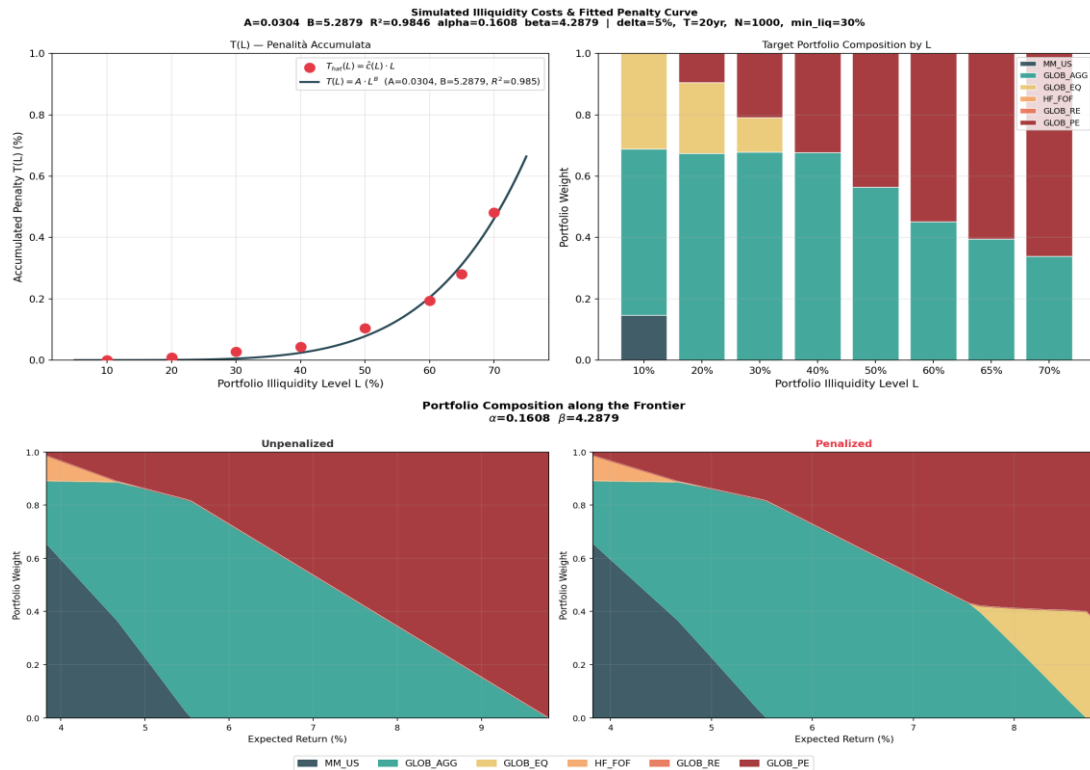
Using estimated α and β in the Penalized MVO

- The **penalized frontier** lies below the unpenalized on a gross return basis but **consistently dominates on a net return basis**, confirming that the unpenalized portfolio's apparent return advantage is entirely offset by its higher illiquidity cost;
- The **penalty reshapes portfolio composition** by systematically substituting PE with GLOB_EQ, reflecting the optimizer's preference for assets that balance return potential against illiquidity cost rather than maximizing gross return through unconstrained PE exposure.



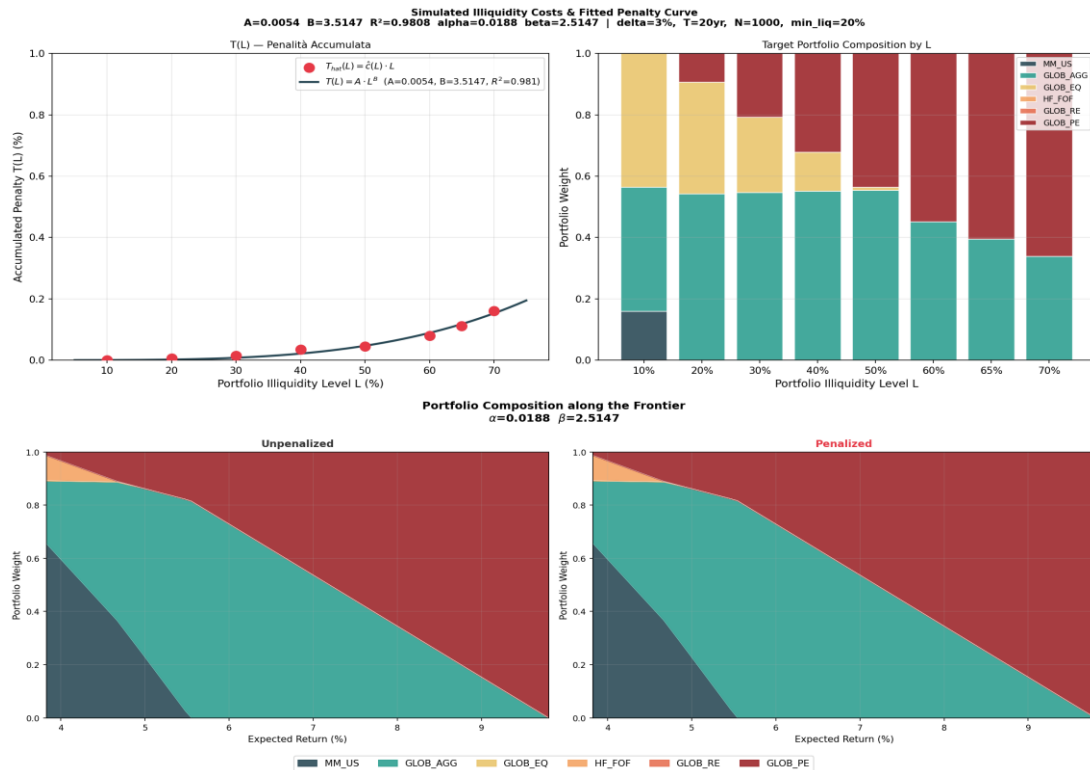
A different Institutional Profile

Fitting and Portfolio Composition of an Endowment ($\alpha = 0.16$, $\beta = 4.28$)



A different Institutional Profile

Fitting and Portfolio Composition of a Family Office ($\alpha = 0.16$, $\beta = 4.28$)



What's next?

Towards a More Realistic and Investor-Specific Calibration Framework

1. A richer simulation framework: introduce liability-aware dynamics into the simulation, making payout requirements a function of the institution's liability duration and funding ratio, rather than a fixed annual rate δ ;
2. Market-implied curve fitting: should empirical data become available on illiquidity costs at different portfolio illiquidity levels, fit $c(L)$ directly on observed costs.